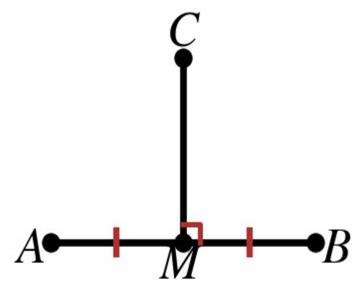
SC Honors Geometry - Nappier
6.1 – Perpendicular and Angle Bisectors • Homework

Name: ___________________________________

Use the Perpendicular Bisector Theorem and its converse. (Figure at right for 1–3.)

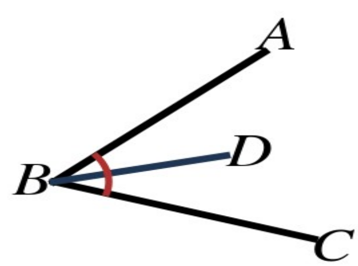


1. $\overline{CM}$ is the perpendicular bisector of $\overline{AB}$. CA = 4x − 1 and CB = 2x + 11. Find x and CA.

2. $\overline{CM}$ is the perpendicular bisector of $\overline{AB}$. AM = 3y + 2 and MB = 5y − 8. Find y and AB.

3. CA = 13 and AM = 5. Find CM.

Use the Angle Bisector Theorem and its converse. (Figure at right for 4.)



4. $\overline{BD}$ bisects ∠ABC. m∠ABD = (5x − 4)° and m∠DBC = (3x + 12)°. Find x and m∠ABC.

5. Point W is in the interior of ∠XYZ and lies on its bisector. Its distances to the two sides are 7z − 9 and 4z + 6. Find z and the common distance.

Write an equation of the perpendicular bisector of the given segment.

6. A(−4, 2) and B(2, 6)

7. C(1, −3) and D(7, 5)

Answer each question. Name the theorem you used.

8. Point Q is 9 cm from R and 9 cm from S. What can you conclude?

9. Point P is in the interior of ∠J and is 5 in. from each side of the angle. What can you conclude?

10. A surveyor must set a marker that is the same distance from two property corners. Describe every location that works.